

Exercise 2.1.3 Use the result of Exercise 2.1.2 to show that if γ is non-null, then $u = As + B$ for constants A and B , $A \neq 0$.

Solution:

From Exercise 2.1.2, the length of $\dot{x}^a = \frac{dx^a}{du}$ is a constant, A .

γ non-null

$$\Rightarrow 0 \neq A^2 = \pm g_{ab} \dot{x}^a \dot{x}^b = \pm g_{ab} \frac{dx^a}{du} \frac{dx^b}{du} = \pm g_{ab} \frac{dx^a}{ds} \frac{dx^b}{ds} \left(\frac{ds}{du} \right)^2.$$

From equation (1.83), the infinitesimal length of the line element is

$$ds^2 = |g_{ab} dx^a dx^b|.$$

So,

$$\pm g_{ab} \frac{dx^a}{ds} \frac{dx^b}{ds} = \pm 1$$

$$\Rightarrow \left(\frac{ds}{du} \right)^2 = A^2 \Rightarrow \frac{ds}{du} = A \Rightarrow s = Au + B \quad \blacksquare$$